A Hardware-Aware Hybrid PQC–QKD Framework for Secure Internet of Medical Things (IoMT)

Submitted by:

Pullabhotla Vamsi Krishna
Register Number: 23BCE1159

Contents

1 Literature Review **2**

1.1 The Computational & Energy Bottlenecks of End-to-End PQC 2

1.2 The Hardware Infeasibility of Pure QKD on the Edge 2

1.3 The Identified Literature Gap 3

2 Proposed Problem Statement **4**

2.1 Project Objectives 4

3 Methodology and Implementation **5**

3.1 Clinical Data Sourcing 5

3.2 Post-Quantum Cryptographic Integration 5

3.3 IoT Hardware Emulation 5

3.4 Implementation Output 5

4 Implementation Results and Scenario Testing **6**

4.1 Scenario 1: Normal Operation (Baseline) 6

4.2 Scenario 2: Active Man-in-the-Middle (MitM) Attack 7

4.3 Scenario 3: Patient Wandering (Signal Degradation) 7

1 Literature Review

The proliferation of the Internet of Medical Things (IoMT) has revolutionized continuous patient monitoring. However, the reliance on classical public-key infrastructure (such as RSA and ECC) renders these networks critically vulnerable to the imminent threat of cryptographically relevant quantum computers (CRQCs). Adversaries are already employing "Harvest Now, Decrypt Later" (HNDL) strategies to capture encrypted clinical telemetry. In reviewing the recent 2025–2026 IEEE Transactions literature, security architectures generally bifurcate into two independent quantum-resistant paradigms: Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD). A critical analysis reveals that applying either paradigm end-to-end in an IoMT environment presents severe physiological and hardware constraints.

1.1 The Computational & Energy Bottlenecks of End-to-End PQC

Post-Quantum Cryptography relies on mathematically complex, quantum-resistant algorithms, predominantly Lattice-Based Cryptography (LBC). While NIST recently standardized ML-KEM (Kyber) based on the Module Learning with Errors (MLWE) problem, its direct application to edge-constrained wearables is highly problematic.

- **Latency and Processing Overhead:** Cao et al. (2025) evaluated the deployment of lattice-based KEMs on constrained microcontrollers (e.g., ARM Cortex-M4 and ESP32). Their empirical benchmarking demonstrated that polynomial multiplication and discrete Gaussian sampling required for PQC encapsulation induce severe latency spikes. Specifically, executing pure PQC took up to **570.3 ms for encapsulation** and **186.7 ms for decapsulation**. For critical IoMT devices like cardiac pacemakers, the Association for the Advancement of Medical Instrumentation (AAMI) mandates a telemetry delay of < 250 ms; thus, end-to-end PQC introduces fatal latency violations.

- **Energy Depletion:** Furthermore, Saeed et al. (2025) and Sharma & Rani (2025) highlighted the energy tax of high-dimensional matrix operations. Their findings indicate that PQC encapsulation consumes approximately **2.4 mJ to 3.1 mJ** per transaction. On a continuous 360 Hz ECG monitor, this computational payload causes rapid battery exhaustion, necessitating frequent, invasive battery replacements for implanted medical devices.

1.2 The Hardware Infeasibility of Pure QKD on the Edge

Conversely, Quantum Key Distribution (QKD) provides unconditional, information-theoretic security governed by the laws of quantum mechanics, utilizing protocols such as BB84 and E91.

- **Physical Infrastructure Constraints:** Zheng et al. (2025) and Wang et al. (2026) successfully demonstrated QKD's efficacy in optimizing key routing over wide-area networks. By monitoring the Quantum Bit Error Rate (QBER), these systems can detect eavesdroppers with 100% mathematical certainty. However, QKD requires dedicated physical infrastructure, including single-photon avalanche diodes (SPADs), phase modulators, and an uninterrupted optical fiber or line-of-sight free-space optical (FSO) link.

- **The Edge Barrier:** It is physically and practically impossible to integrate optical fiber transceivers or maintain line-of-sight lasers on a patient's mobile, battery-operated wearable device. Consequently, pure QKD is strictly relegated to stationary, high-resource backbone networks.

1.3 The Identified Literature Gap

The state-of-the-art literature currently treats PQC and QKD as mutually exclusive solutions. PQC is entirely software-based but computationally paralyzing for IoT, while QKD offers perfect security but poses insurmountable hardware barriers for edge devices. There is a distinct absence of **tiered, hybrid architectural models** that strategically map these cryptographic domains to their respective hardware-appropriate network layers.

To date, no peer-reviewed framework has successfully demonstrated a cryptographic translation gateway that bridges short-range, computationally restricted PQC (Wearable-to-Edge) with long-range, information-theoretically secure QKD (Edge-to-Datacenter) while maintaining clinical real-time latency bounds.

2 Proposed Problem Statement

In response to the identified architectural vulnerabilities and physical constraints of existing quantum-resistant proposals, this project defines the following problem statement:

> *"To design, simulate, and empirically validate a tiered* ***Hybrid PQC–QKD Edge Computing Framework for the Internet of Medical Things (IoMT)****. This architecture aims to resolve the quantum computational bottleneck on low-resource edge devices by confining lightweight Post-Quantum Cryptography (NIST ML-KEM) to short-range Wearable-to-Edge transmissions, while offloading high-bandwidth backbone security to a simulated Quantum Key Distribution (BB84) channel. The primary goal is to prove that this hybrid cryptographic translation model can maintain end-to-end quantum resistance without violating the strict real-time clinical safety threshold of* < 250 *ms."*

2.1 Project Objectives

To systematically address this problem, the project is divided into four primary objectives:

1. **Tier 1 - Wearable PQC Emulation:** To implement and evaluate the NIST FIPS 203 standardized ML-KEM-512 algorithm on a simulated constrained IoMT node. This includes injecting hardware-specific scaling delays to accurately model execution on an ESP32 microcontroller.

2. **Tier 2 - Hybrid Cryptographic Edge Gateway:** To engineer an edge computing node that acts as a secure translator. This gateway must decapsulate incoming PQC payloads, verify data integrity via AES-GCM, and immediately re-encrypt the clinical payload using a QKD-derived symmetric key pool.

3. **Tier 3 - QKD Optical Backbone Simulation:** To model a Quantum Key Distribution link using the BB84 protocol via IBM Qiskit. This involves simulating quantum state preparation, basis sifting, and Quantum Bit Error Rate (QBER) monitoring to establish a perfectly secure transmission corridor to the datacenter.

4. **Clinical Data Ingestion & Benchmarking:** To validate the system's physiological relevance by ingesting and streaming real-world electrocardiogram (ECG) telemetry from the PhysioNet MIT-BIH Arrhythmia Database, and comparing the framework's latency and energy consumption against legacy RSA/ECC baselines.

3 Methodology and Implementation

For the 30% implementation milestone, the foundational Tier 1 (Wearable Client) and Tier 2 (Edge Gateway) components of the proposed hybrid architecture have been developed. The focus of this phase was to establish a quantum-resistant, hardware-aware communication link over the initial Wearable-to-Edge hop.

3.1 Clinical Data Sourcing

To validate the system using realistic physiological parameters, the implementation integrates the Python `wfdb` library to fetch real patient telemetry. Specifically, electrocardiogram (ECG) lead MLII millivolt readings are extracted from the PhysioNet MIT-BIH Arrhythmia Database (Record 100). This data is dynamically parsed and streamed to serve as the baseline medical payload.

3.2 Post-Quantum Cryptographic Integration

The security pipeline was developed utilizing the Open Quantum Safe (`liboqs`) Python wrapper. The Wearable node is programmed to encapsulate a shared secret using the NIST-standardized ML-KEM-512 algorithm. This encapsulated secret is subsequently utilized to derive an AES-256-GCM key. The ECG payload is symmetrically encrypted and authenticated with a generated tag, ensuring both data confidentiality and integrity during transmission to the Edge Gateway.

3.3 IoT Hardware Emulation

Running lattice-based cryptography on a high-performance PC does not reflect IoMT realities. Therefore, a deterministic hardware delay scaling module was implemented. The simulation measures the base algorithmic execution time and applies a predefined multiplier to emulate specific hardware constraints. The Wearable node is scaled to emulate a 240 MHz ESP32 microcontroller (yielding realistic PQC latencies of $\approx$ 25–35 ms), while the Edge node is scaled to emulate a standard hospital NUC router ($\approx$ 1.5 ms).

3.4 Implementation Output

The operational state of the 30% framework is visualized through a custom dual-pane Terminal User Interface (TUI) developed with the `rich` library. The console actively logs the data stream, displaying packet sequence numbers, raw ECG voltages, hardware-emulated PQC latencies, and the final AES-GCM verification status at the edge receiver.

4 Implementation Results and Scenario Testing

To rigorously validate the Tier 1 and Tier 2 architecture, the simulation framework includes a real-time Physics and Network Engine. The system was subjected to three distinct operational scenarios, visualized through the custom Terminal User Interface (TUI). The results confirm the framework's resilience and accurate cryptographic handling under both stable and adversarial conditions.

4.1 Scenario 1: Normal Operation (Baseline)

In the baseline scenario, three IoMT nodes (streaming High-Frequency ECG, Apnea, and Sleep telemetry) operate within optimal physical proximity to the Edge Gateway.

- **Observation:** As depicted in Figure 1, the nodes maintain stable Received Signal Strength Indicators (RSSI) between -47 dBm and -59 dBm.
- **Cryptographic Validation:** The Edge Gateway successfully decapsulates the ML-KEM-512 shared secrets and verifies the AES-GCM tags. The raw clinical payloads (e.g., MLII/V5 readings and Blood Pressure) are accurately decrypted with a 0% tampering rate, proving the nominal efficiency of the hybrid pipeline.

Figure 1: Dashboard view during **Scenario 1 (Normal Operation)**. The network topology reflects stable RSSI values, and the log confirms continuous, verified packet reception without cryptographic failures.

4.2 Scenario 2: Active Man-in-the-Middle (MitM) Attack

To test data integrity, an adversary (Eve) is simulated attempting to intercept and modify the ciphertext transmitted from Node-02 (DEV-2) before it reaches the Edge Gateway.

- **Observation:** Because the adversary lacks the ML-KEM-512 private key required to properly decapsulate and re-encrypt the AES-GCM symmetric key, the cryptographic tag check fails upon arrival at the Edge node.

- **System Response:** The framework correctly flags the modified packets, shifting Node-02's status to *MALICIOUS TAMPER DETECTED* and immediately dropping the corrupted telemetry to prevent clinical misdiagnosis.

Figure 2: Dashboard view during **Scenario 2 (MitM Attack)**. The system successfully detects the cryptographic integrity violation on DEV-2 and halts the injection of tampered data.

4.3 Scenario 3: Patient Wandering (Signal Degradation)

To test the physical networking constraints, a "Patient Wandering" event is triggered, rapidly increasing the physical distance between Node-03 (DEV-3) and the Edge Gateway.

- **Observation:** As the patient moves out of range, the RSSI drops precipitously below the operational threshold (≈ -90 dBm).

- **System Response:** The physics engine simulates real-world wireless constraints, resulting in packet loss. The system properly registers this as a natural environmental failure (*DROPPED - Signal Lost*) rather than a cryptographic attack, allowing the Edge node to safely pause processing for that specific device until the patient returns to the ward.

Figure 3: Dashboard view during **Scenario 3 (Patient Wandering)**. The physics engine accurately models signal attenuation, dropping packets from DEV-3 as it moves beyond the gateway's effective operational radius.